

Discovering activation functions

INTRODUCTION TO DEEP LEARNING WITH PYTORCH

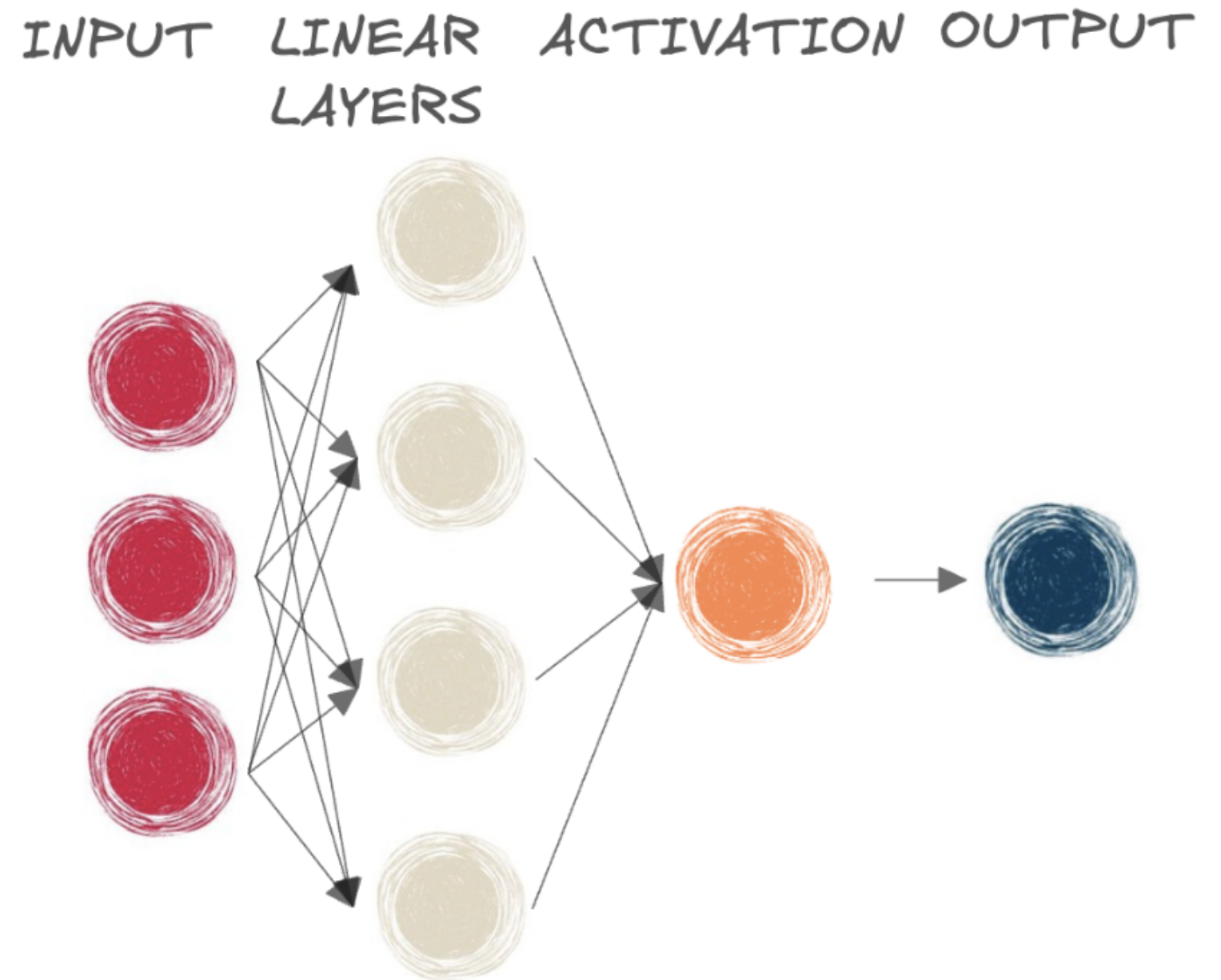


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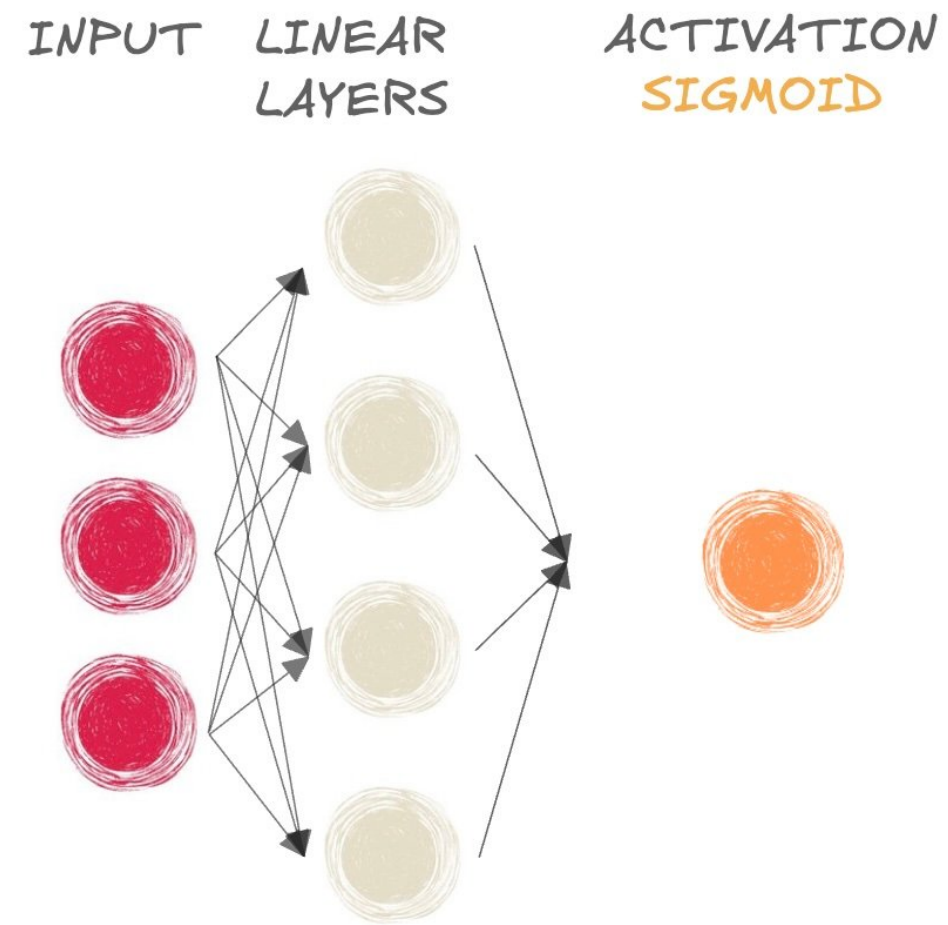
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Activation functions

- **Activation functions** add non-linearity to the network
 - **Sigmoid** for binary classification
 - **Softmax** for multi-class classification
- A network can learn more **complex** relationships with non-linearity
- "Pre-activation" output passed to the activation function



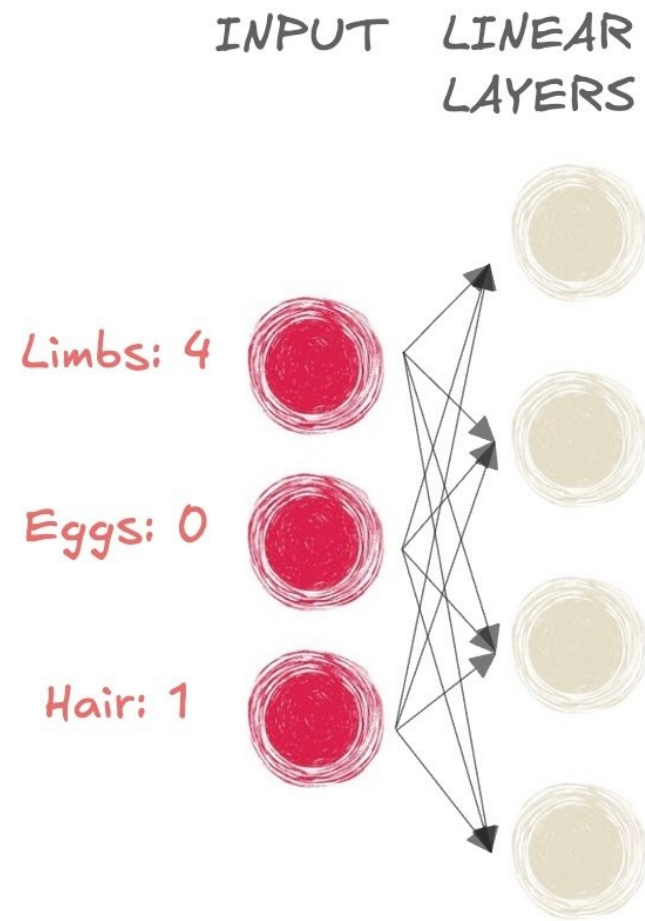
Meet the sigmoid function



- Mammal or not?



Meet the sigmoid function

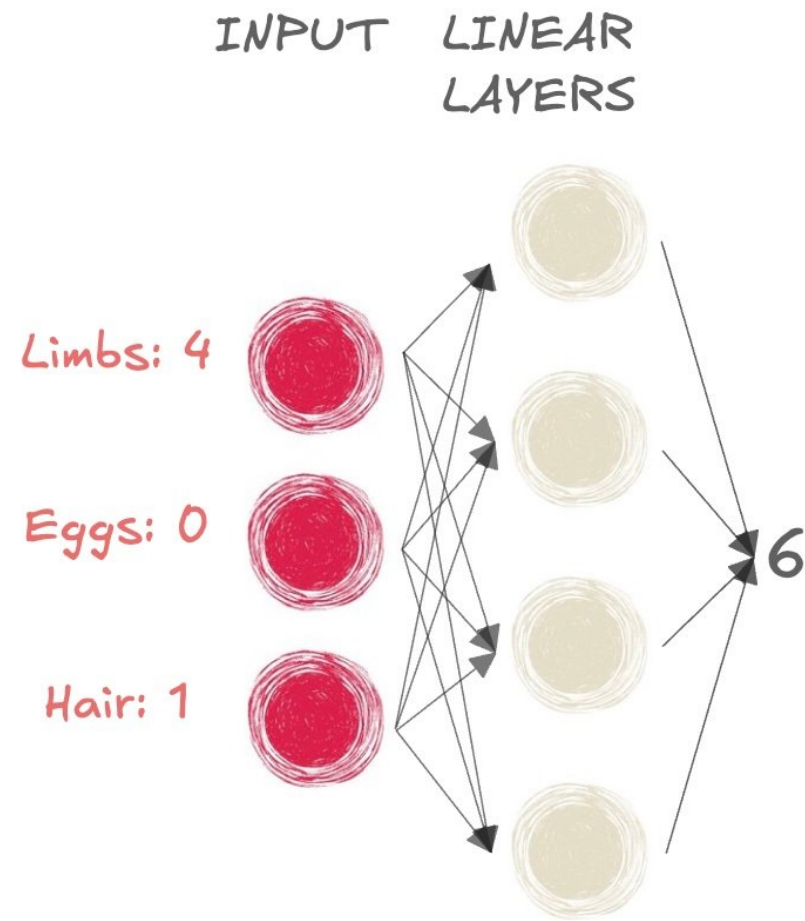


- Mammal or not?



- Input:
 - Limbs: 4
 - Eggs: 0
 - Hair: 1

Meet the sigmoid function

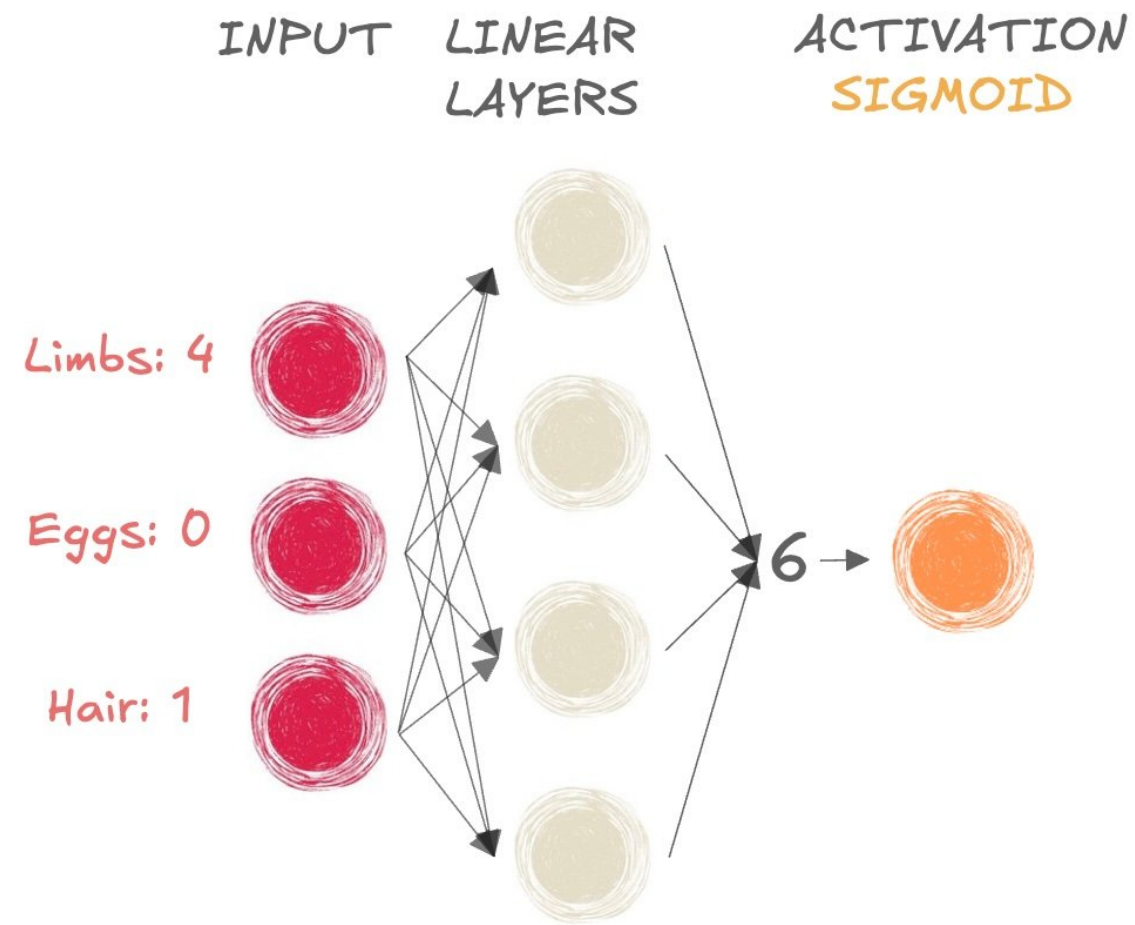


- Mammal or not?



- Output to the linear layers is 6

Meet the sigmoid function

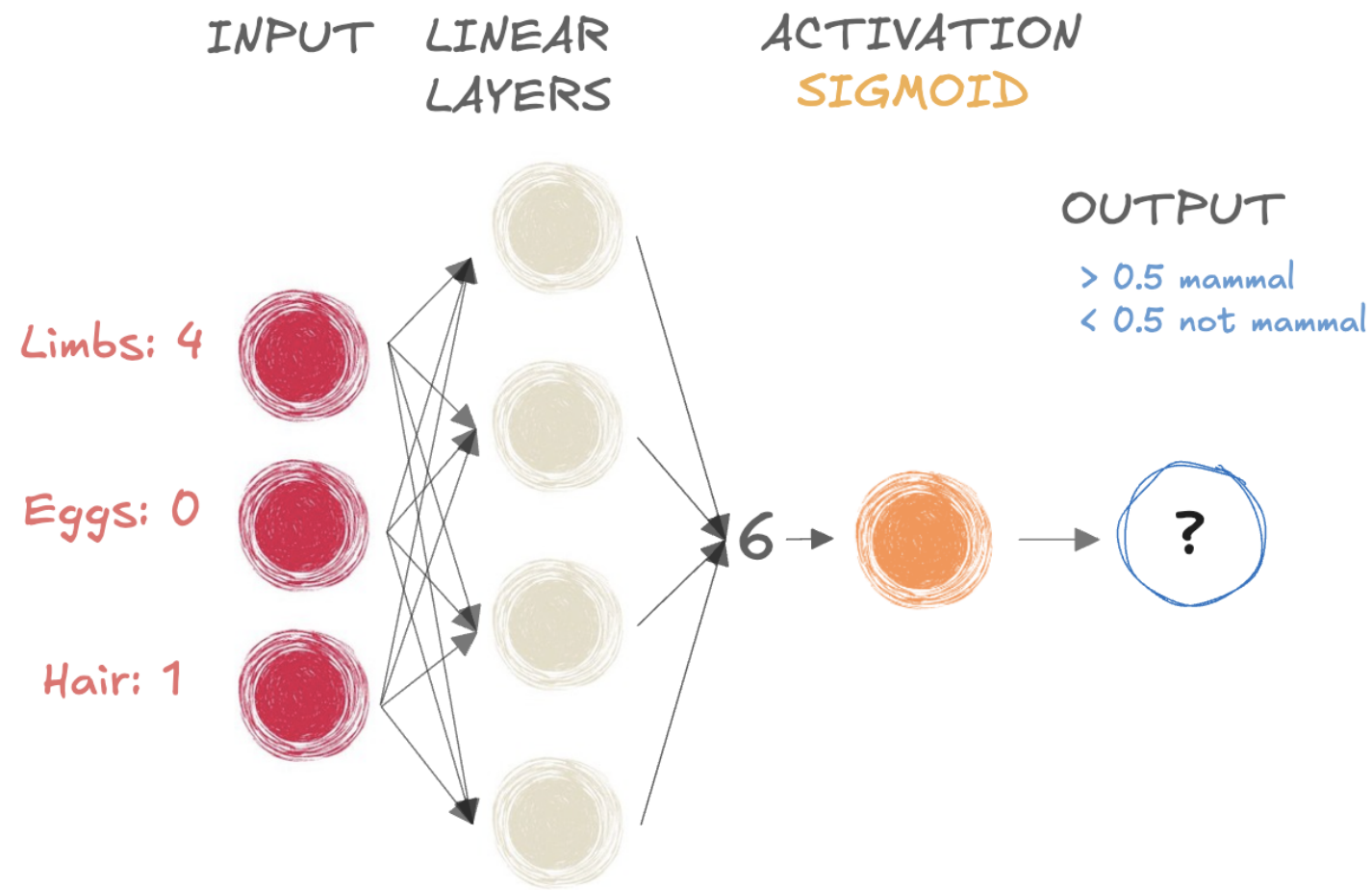


- Mammal or not?



- We take the pre-activation output (6) and pass it to the sigmoid function

Meet the sigmoid function



- Mammal or not?



- We take the pre-activation output (6) and pass it to the sigmoid function
- Obtain a value between 0 and 1
- If output is > 0.5 , class label = 1 (mammal)
- If output is ≤ 0.5 , class label = 0 (not mammal)

Meet the sigmoid function

```
import torch
import torch.nn as nn

input_tensor = torch.tensor([[6]])
sigmoid = nn.Sigmoid()
output = sigmoid(input_tensor)
print(output)
```

```
tensor([[0.9975]])
```


Activation as the last layer

```
model = nn.Sequential(  
    nn.Linear(6, 4), # First linear layer  
    nn.Linear(4, 1), # Second linear layer  
    nn.Sigmoid() # Sigmoid activation function  
)
```

Sigmoid as last step in network of linear layers is **equivalent** to traditional logistic regression

Getting acquainted with softmax

- Three classes:

Getting acquainted with softmax

- Three classes:



BIRD (0)

Getting acquainted with softmax

- Three classes:



BIRD (0)



MAMMAL (1)

Getting acquainted with softmax

- Three classes:

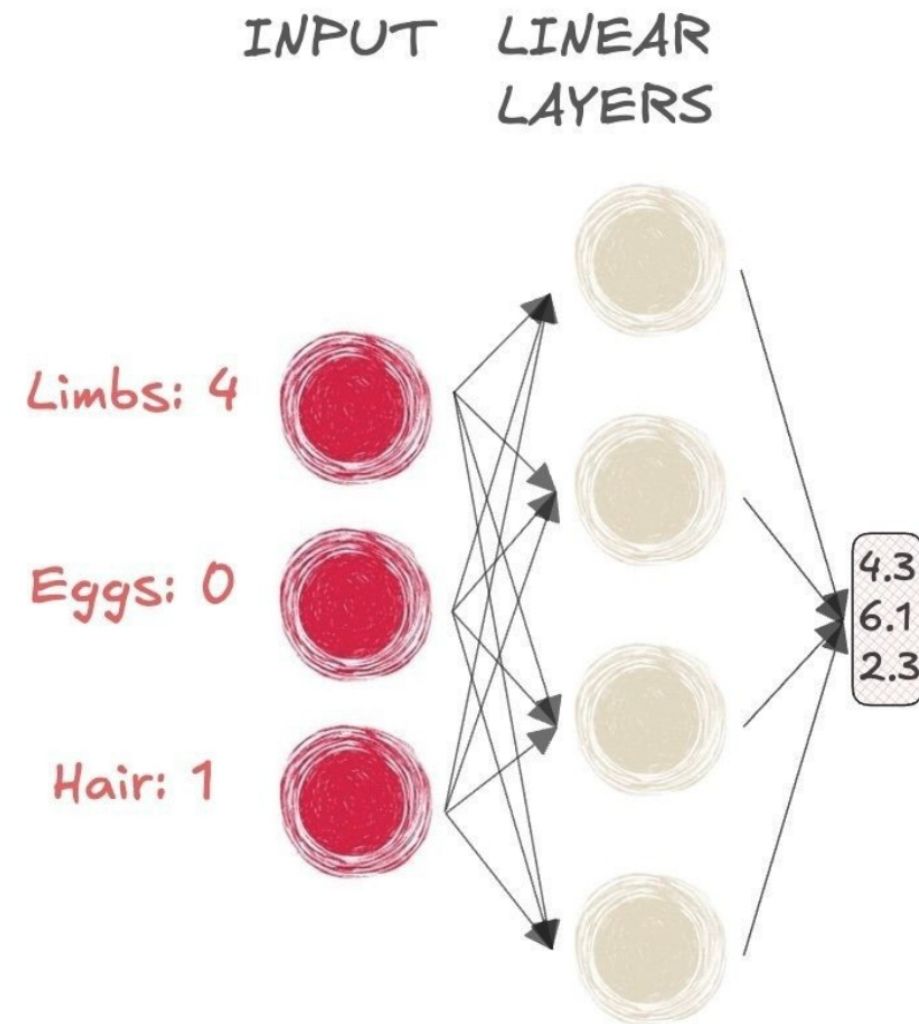


BIRD (0)

MAMMAL (1)

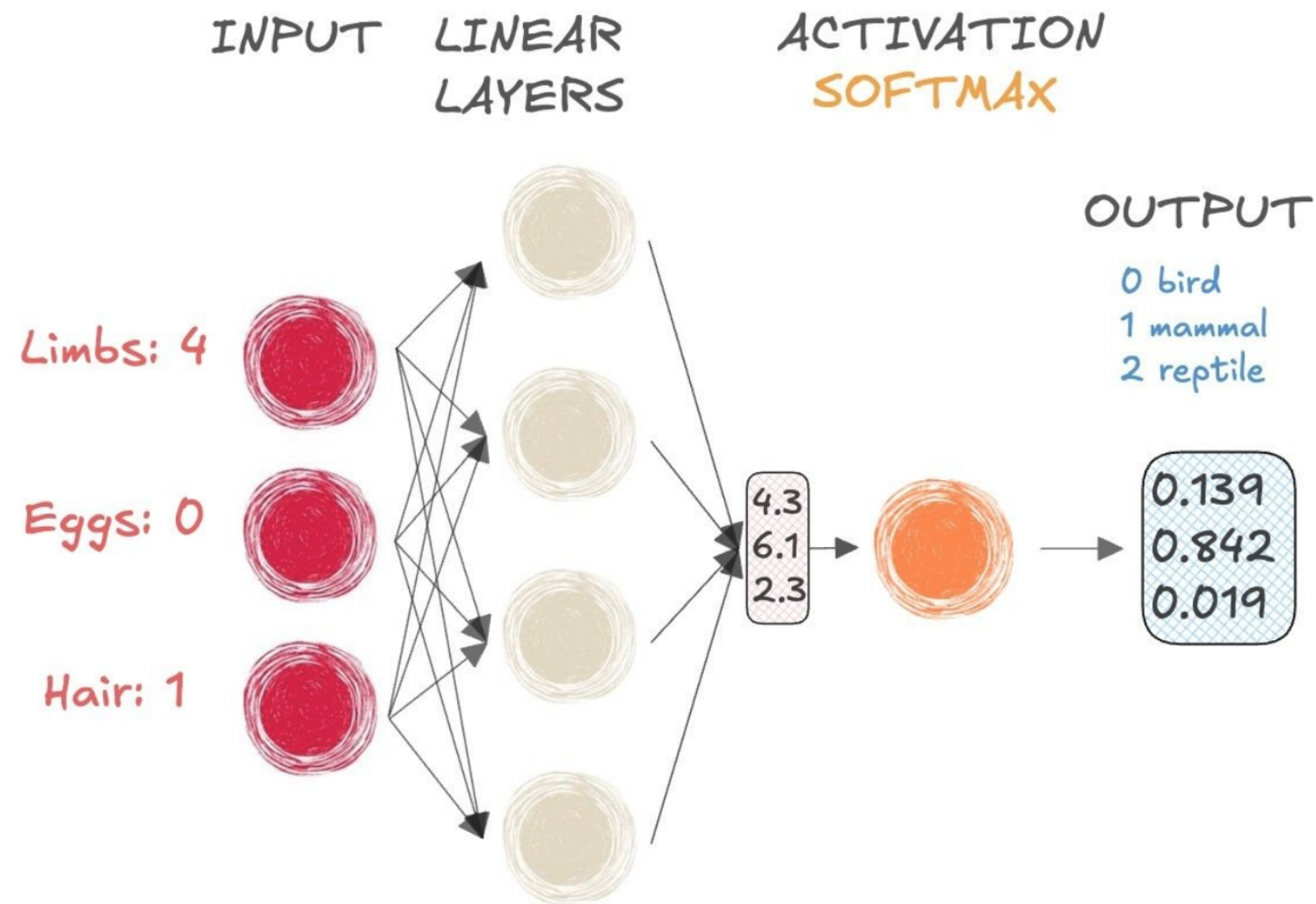
REPTILE (2)

Getting acquainted with softmax



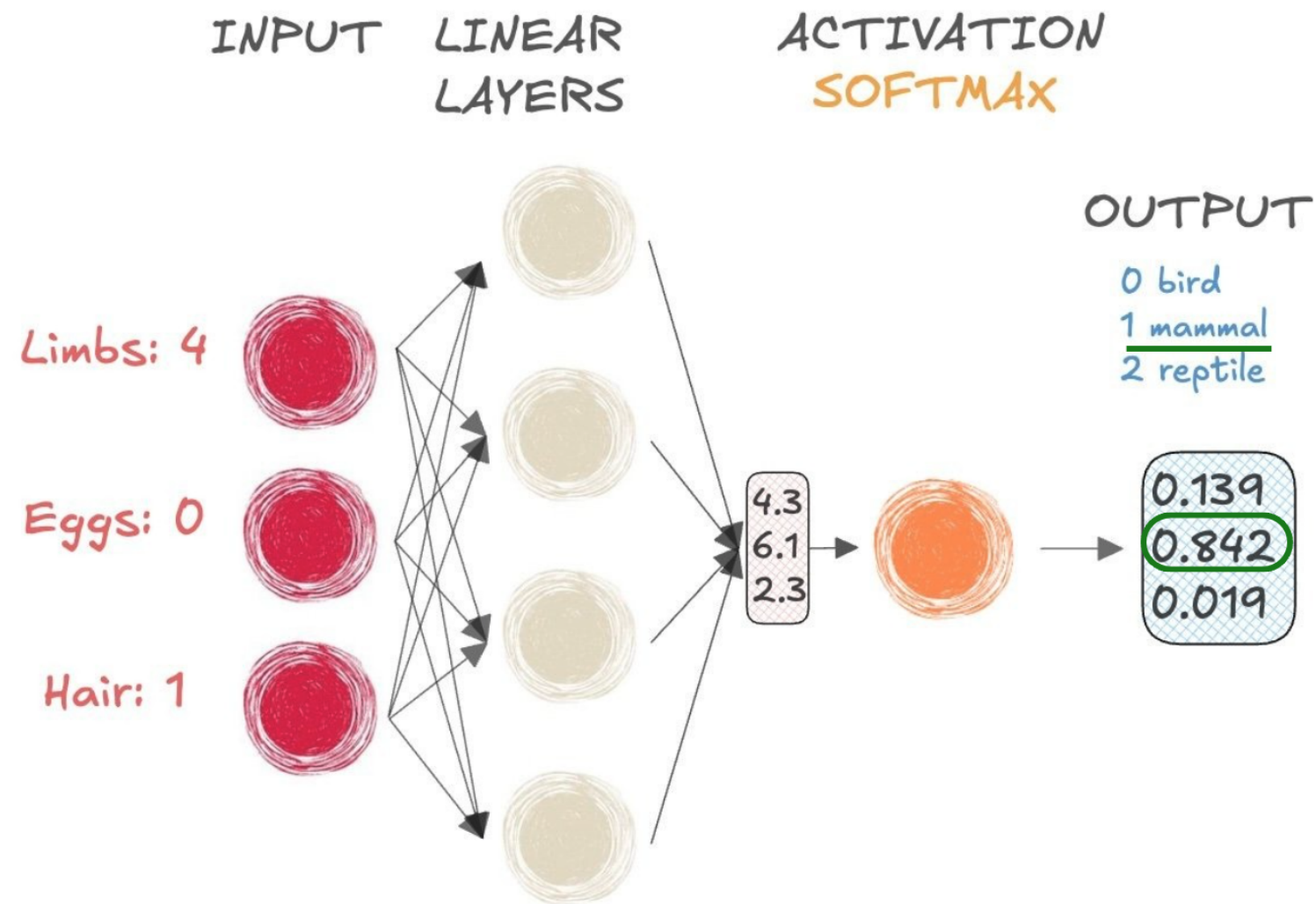
- Takes three-dimensional as input and outputs the same shape

Getting acquainted with softmax



- Takes three-dimensional as input and outputs the same shape
- Outputs a probability distribution:
 - Each element is a probability (it's bounded between 0 and 1)
 - The sum of the output vector is equal to 1

Getting acquainted with softmax



- Takes three-dimensional as input and outputs the same shape
- Outputs a probability distribution:
 - Each element is a probability (it's bounded between 0 and 1)
 - The sum of the output vector is equal to 1

Getting acquainted with softmax

```
import torch
import torch.nn as nn

# Create an input tensor
input_tensor = torch.tensor(
    [[4.3, 6.1, 2.3]])

# Apply softmax along the last dimension
probabilities = nn.Softmax(dim=-1)
output_tensor = probabilities(input_tensor)
print(output_tensor)
```

```
tensor([[0.1392, 0.8420, 0.0188]])
```

- `dim = -1` indicates softmax is applied to the input tensor's last dimension
- `nn.Softmax()` can be used as last step in `nn.Sequential()`

Let's practice!

INTRODUCTION TO DEEP LEARNING WITH PYTORCH

Running a forward pass

INTRODUCTION TO DEEP LEARNING WITH PYTORCH

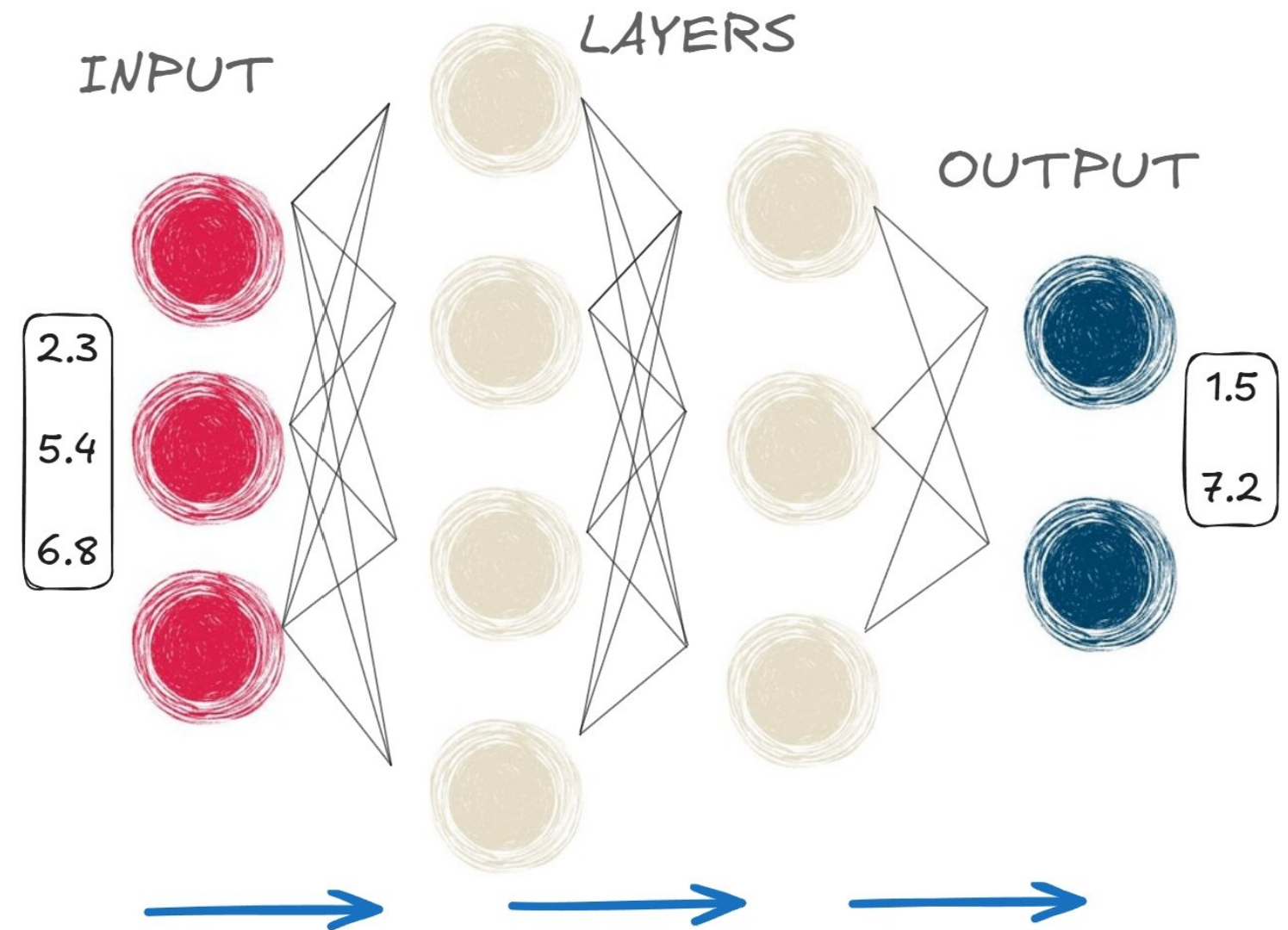


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What is a forward pass?

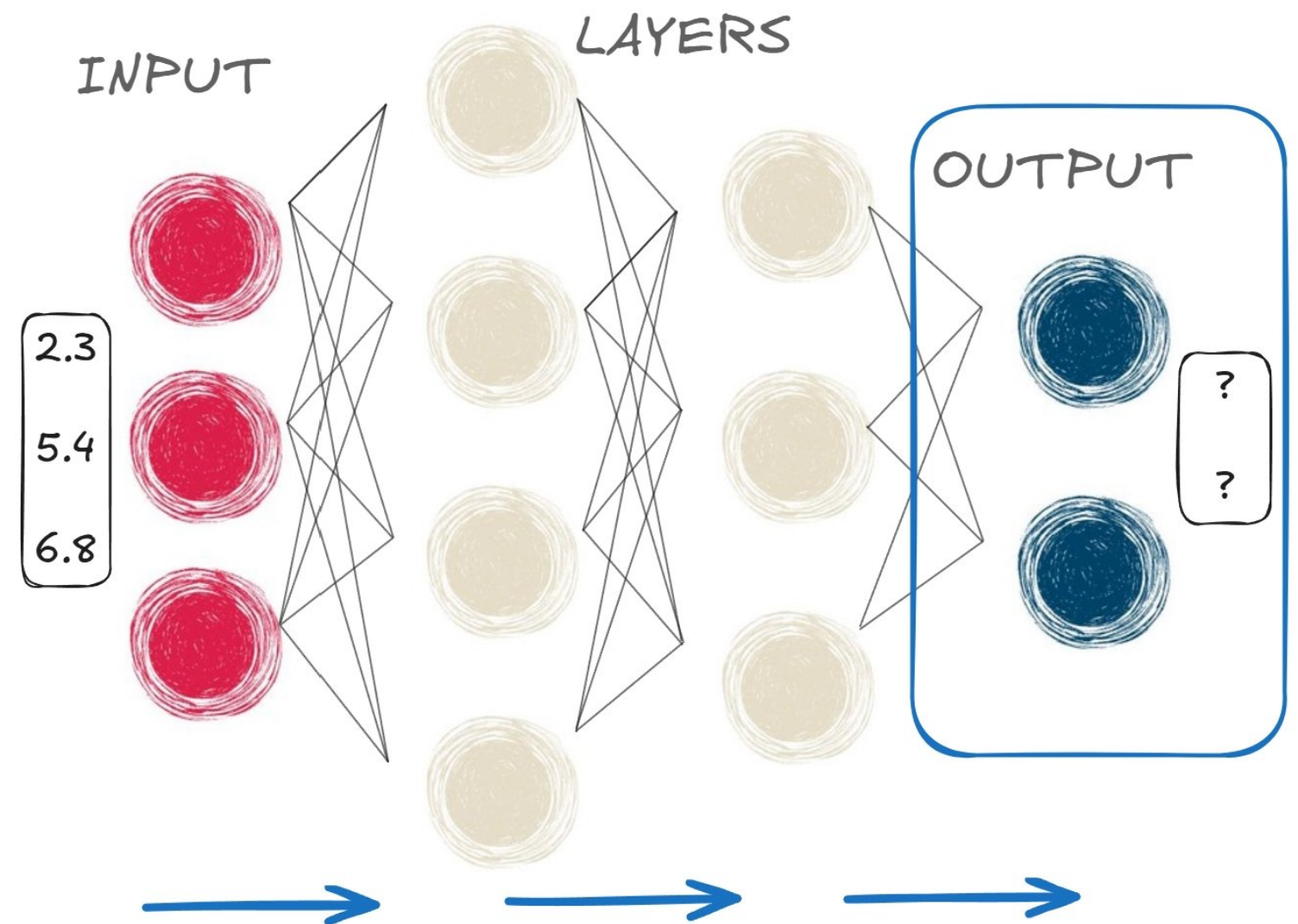
- Input data **flows** through layers
- **Calculations** performed at each layer
- Final layer generates **outputs**
- Outputs produced based on **weights** and **biases**
- Used for **training** and making **predictions**



What is a forward pass?

Possible outputs:

- Binary classification
- Multi-class classification
- Regressions



Binary classification: forward pass

```
# Create input data of shape 5x6
```

```
input_data = torch.tensor(  
    [[-0.4421, 1.5207, 2.0607, -0.3647, 0.4691, 0.0946],  
     [-0.9155, -0.0475, -1.3645, 0.6336, -1.9520, -0.3398],  
     [ 0.7406, 1.6763, -0.8511, 0.2432, 0.1123, -0.0633],  
     [-1.6630, -0.0718, -0.1285, 0.5396, -0.0288, -0.8622],  
     [-0.7413, 1.7920, -0.0883, -0.6685, 0.4745, -0.4245]])
```

5 animals

6 features

```
# Create binary classification model
```

```
model = nn.Sequential(  
    nn.Linear(6, 4), # First linear layer  
    nn.Linear(4, 1), # Second linear layer  
    nn.Sigmoid() # Sigmoid activation function  
)
```

Binary classification: forward pass

```
# Pass input data through model
output = model(input_data)
print(output)
```

```
tensor([[0.5188], [0.3761], [0.5015], [0.3718], [0.4663]],
        grad_fn=<SigmoidBackward0>)
```

- Output: five probabilities between 0 and 1, one for each animal
- Classification (0.5 threshold):
 - Class = 1 (mammal) for values ≥ 0.5 (0.5188 , 0.5015)
 - Class = 0 (not mammal) for values < 0.5 (0.3761 , 0.3718 , 0.4633)

Multi-class classification: forward pass

- Class 1 - mammal, class 2 - bird, class 3 - reptile

```
n_classes = 3

# Create multi-class classification model
model = nn.Sequential(
    nn.Linear(6, 4), # First linear layer
    nn.Linear(4, n_classes), # Second linear layer
    nn.Softmax(dim=-1) # Softmax activation
)

# Pass input data through model
output = model(input_data)
print(output.shape)
```

```
torch.Size([5, 3])
```

Multi-class classification: forward pass

```
print(output)
```

```
tensor([[0.4969, 0.3606, 0.1425],  
        [0.5105, 0.3262, 0.1633],  
        [0.3253, 0.3174, 0.3572],  
        [0.5499, 0.3361, 0.1141],  
        [0.4117, 0.3366, 0.2517]], grad_fn=<SoftmaxBackward0>)
```

5 animals

probabilities for each class

- Each row sums to one
- Predicted label = class with the highest probability
- Row 1 = class 1 (mammal), row 2 = class 1 (mammal), row 3 = class 3 (reptile)

Regression: forward pass

```
# Create regression model
model = nn.Sequential(
    nn.Linear(6, 4), # First linear layer
    nn.Linear(4, 1) # Second linear layer
)

# Pass input data through model
output = model(input_data)

# Return output
print(output)
```

```
tensor([[0.3818],
        [0.0712],
        [0.3376],
        [0.0231],
        [0.0757]],
        grad_fn=<AddmmBackward0>)
```

Let's practice!

INTRODUCTION TO DEEP LEARNING WITH PYTORCH

Using loss functions to assess model predictions

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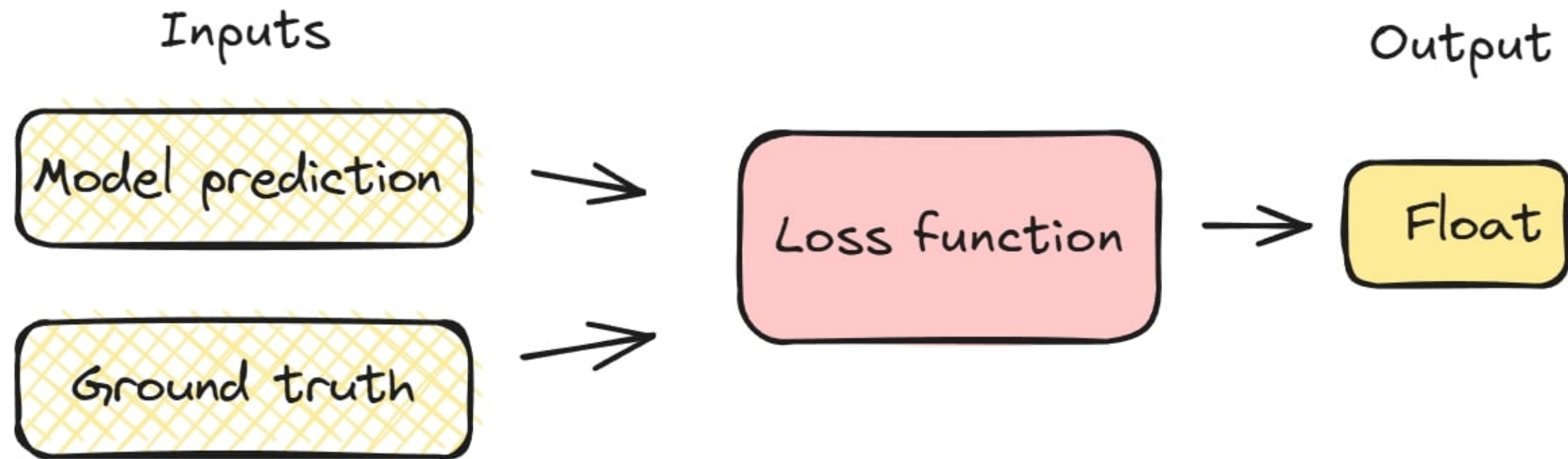
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Why do we need a loss function?

- Tells us how good our model is during training
- Takes a model prediction $\hat{y}$ and ground truth y
- Outputs a float



Why do we need a loss function?

- Class 0 - mammal, class 1 - bird, class 2 - reptile

Hair	Feathers	Eggs	Milk	Fins	Legs	Tail	Domestic	Catsize	Class
1	0	0	1	0	4	0	0	1	0

- Predicted class = 0 -> **correct** = low loss
 - Predicted class = 1 -> **wrong** = high loss
 - Predicted class = 2 -> **wrong** = high loss
-
- Our goal is to **minimize loss**

One-hot encoding concepts

- $loss = F(y, \hat{y})$
- y is a single **integer** (class label)
 - e.g. $y = 0$ when y is a mammal
- $\hat{y}$ is a **tensor** (prediction before softmax)
 - If N is the number of classes, e.g. $N = 3$
 - $\hat{y}$ is a tensor with N dimensions,
 - e.g. $\hat{y} = [-5.2, 4.6, 0.8]$

One-hot encoding concepts

- Convert an integer y to a tensor of zeros and ones

	ground truth $y = 0$ number of classes $N = 3$		
class	0	1	2
one-hot encoding	1	0	0

Transforming labels with one-hot encoding

```
import torch.nn.functional as F
```

```
print(F.one_hot(torch.tensor(0), num_classes = 3))
```

```
tensor([1, 0, 0])
```

```
print(F.one_hot(torch.tensor(1), num_classes = 3))
```

```
tensor([0, 1, 0])
```

```
print(F.one_hot(torch.tensor(2), num_classes = 3))
```

```
tensor([0, 0, 1])
```

Cross entropy loss in PyTorch

```
from torch.nn import CrossEntropyLoss

scores = torch.tensor([-5.2, 4.6, 0.8])
one_hot_target = torch.tensor([1, 0, 0])

criterion = CrossEntropyLoss()
print(criterion(scores.double(), one_hot_target.double()))
```

```
tensor(9.8222, dtype=torch.float64)
```

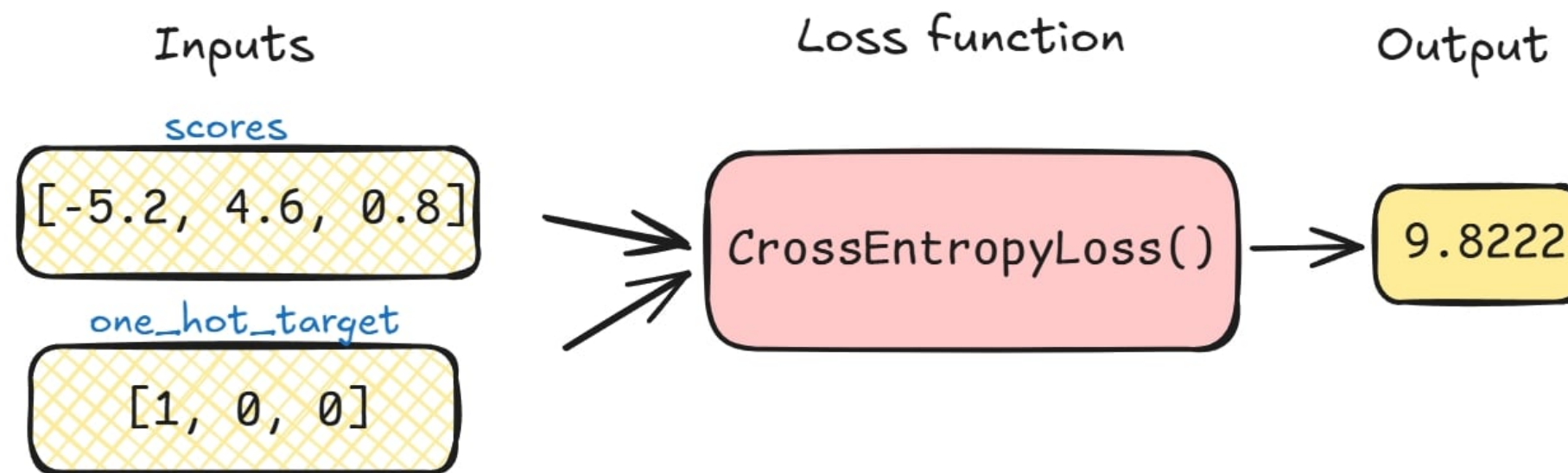
Bringing it all together

Loss function takes:

- **scores** - model predictions **before** the final softmax function
- **one_hot_target** - one hot encoded ground truth label

Loss function outputs:

- **loss** - a single float



Let's practice!

INTRODUCTION TO DEEP LEARNING WITH PYTORCH

Using derivatives to update model parameters

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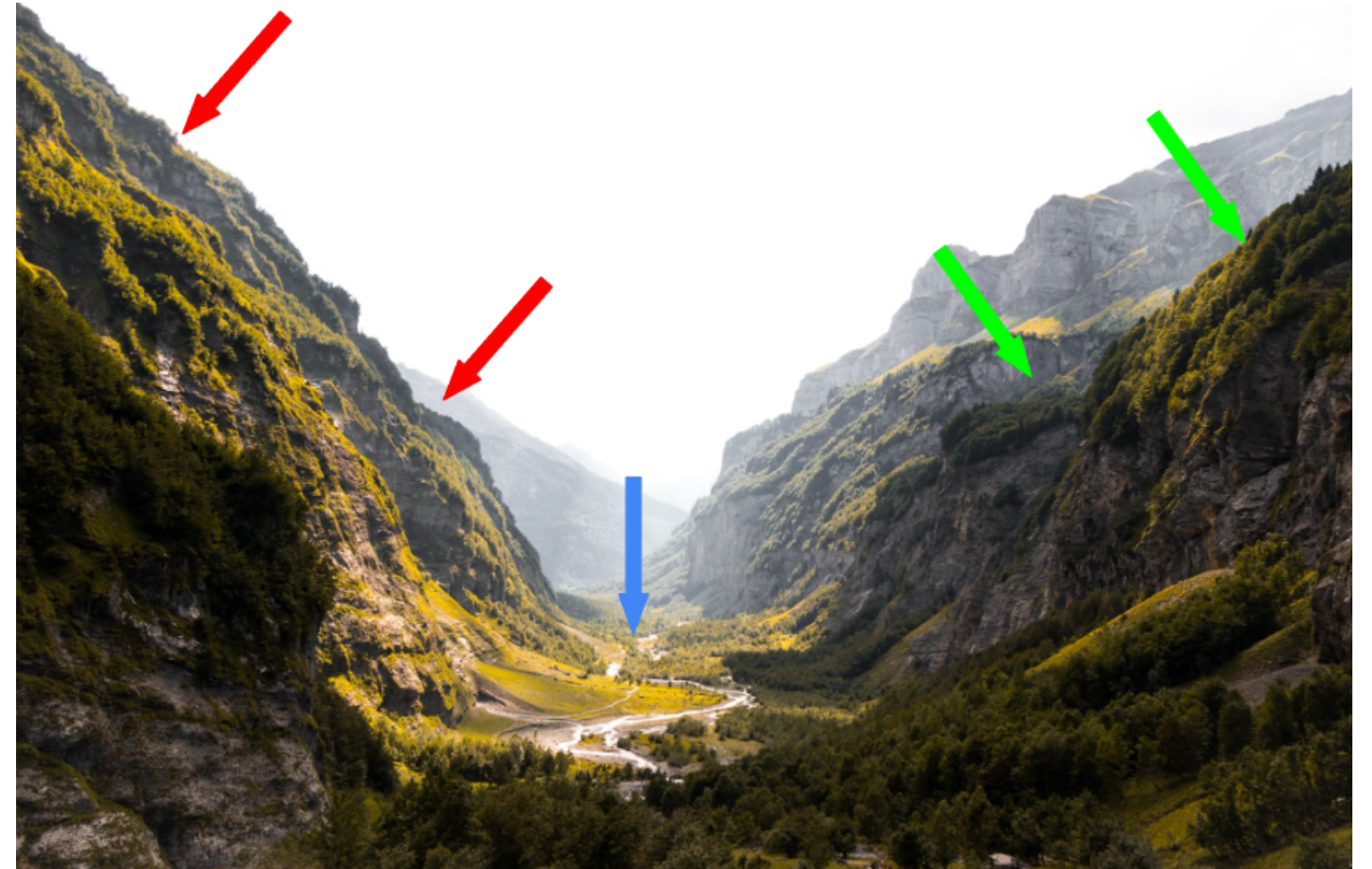
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An analogy for derivatives

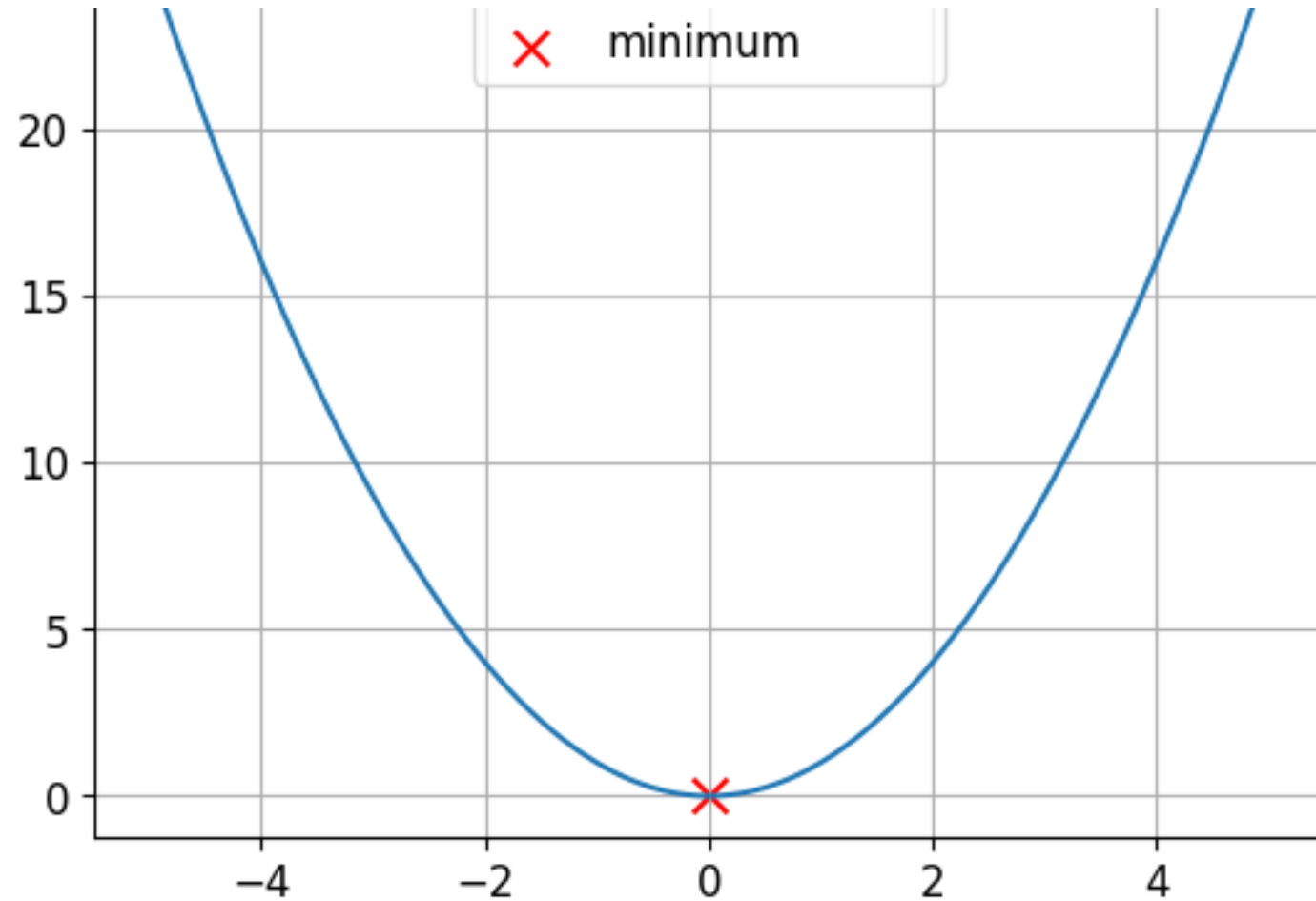
Derivative represents the **slope of the curve**

- **Steep slopes (red arrows):**
 - Large steps, **derivative is high**
- **Gentler slopes (green arrows):**
 - Small steps, **derivative is low**
- **Valley floor (blue arrow):**
 - Flat, **derivative is zero**

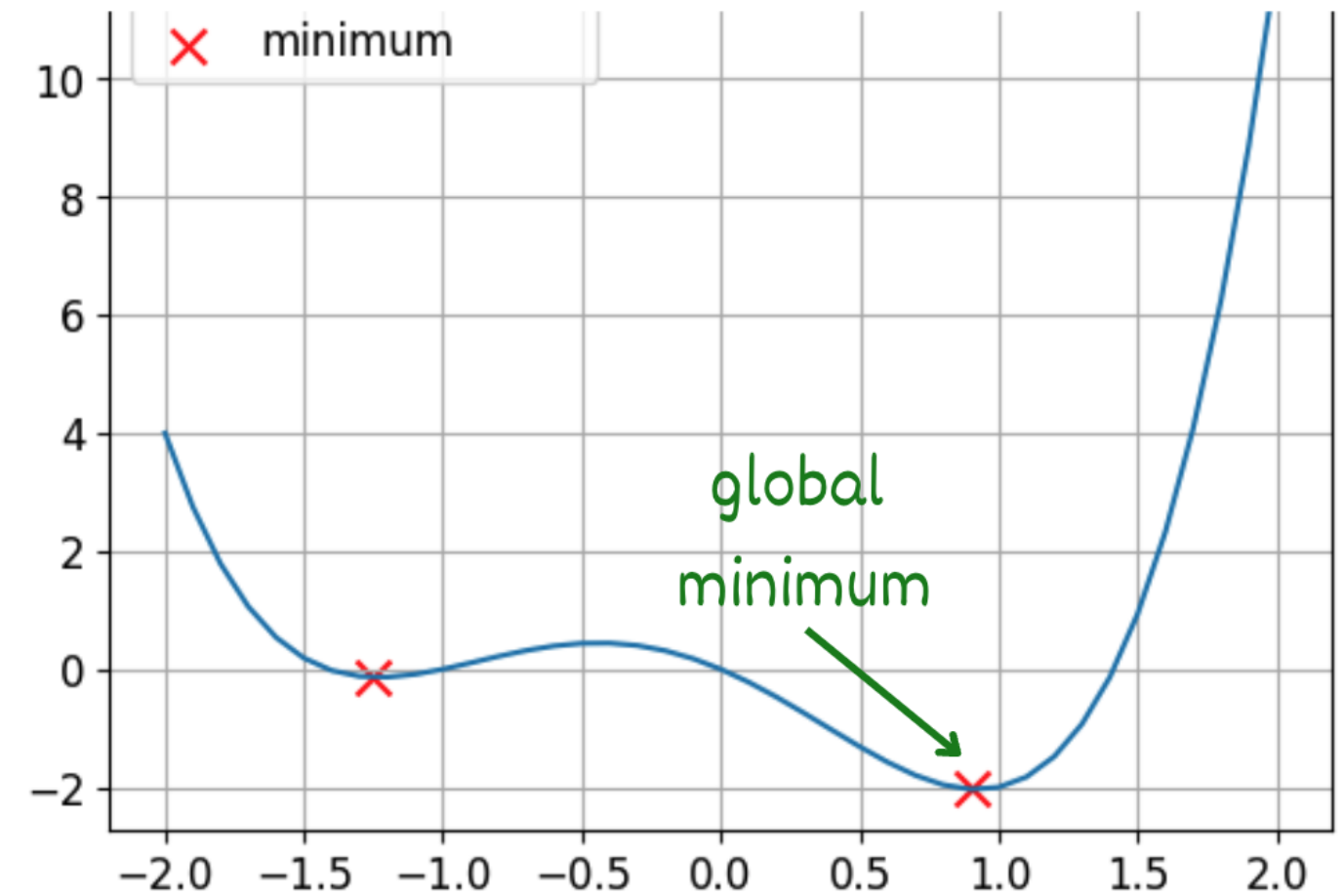


Convex and non-convex functions

This is a convex function

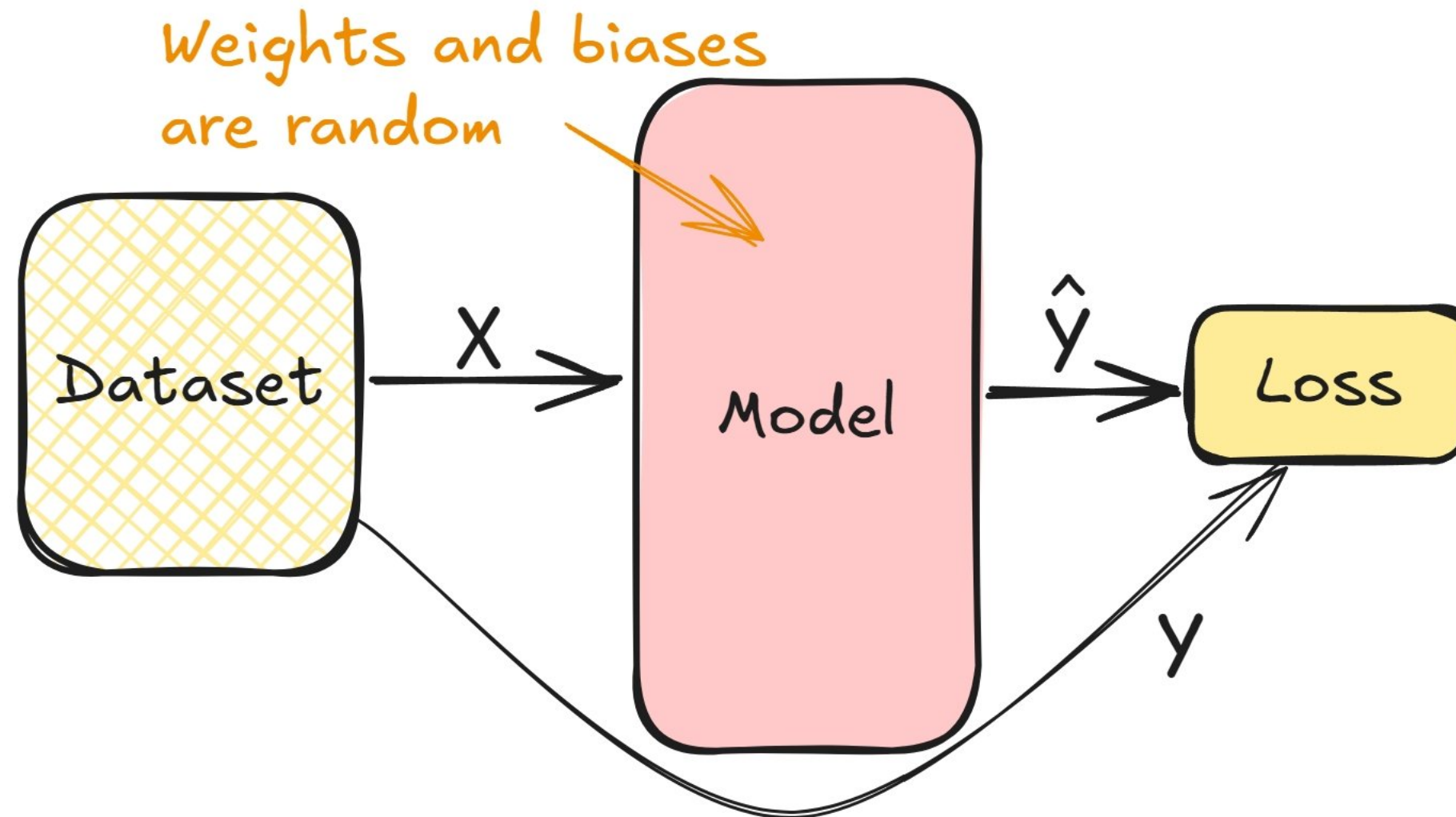


This is a non-convex function



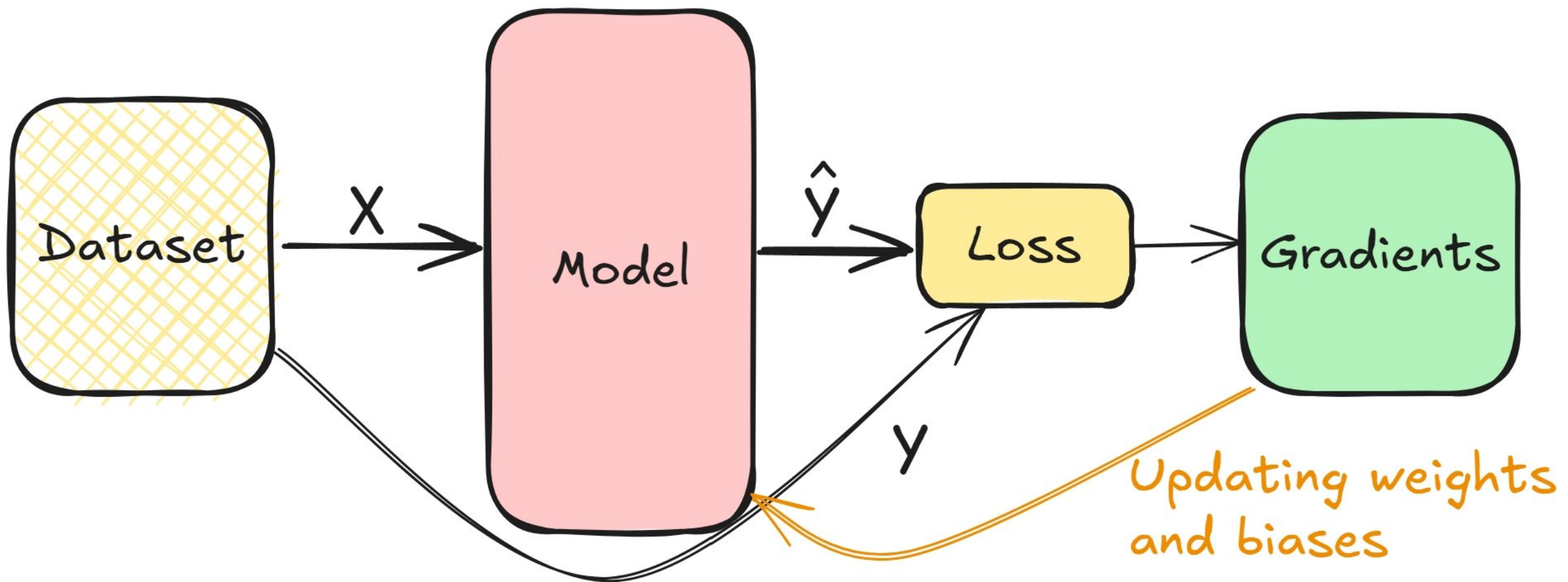
Connecting derivatives and model training

- Compute the loss in the forward pass during training



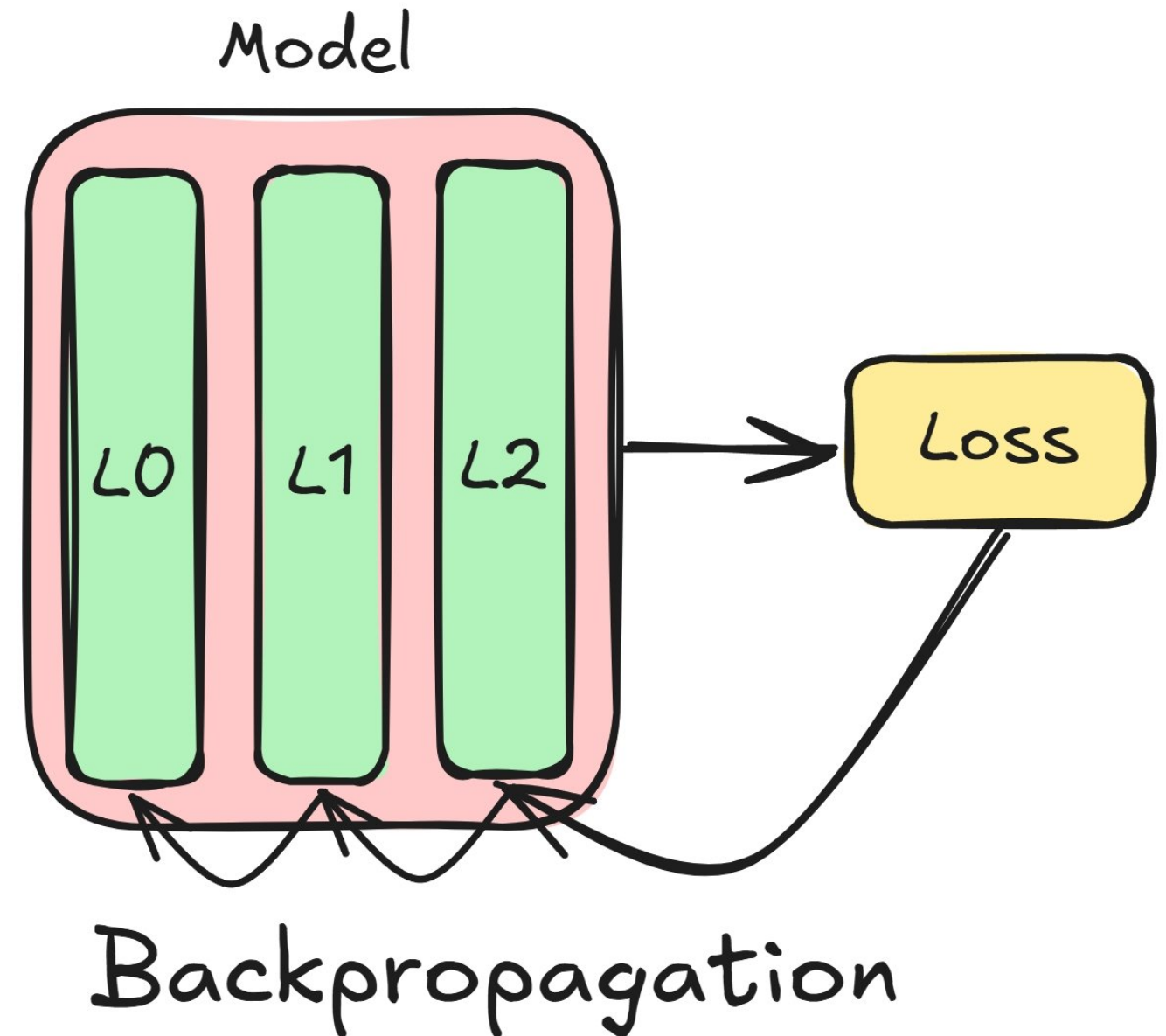
Connecting derivatives and model training

- Gradients help minimize loss, tune layer weights and biases
- Repeat until the layers are tuned



Backpropagation concepts

- Consider a network made of three layers:
 - Begin with loss gradients for $L2$
 - Use $L2$ to compute $L1$ gradients
 - Repeat for all layers ($L1, L0$)



Backpropagation in PyTorch

Run a forward pass

```
model = nn.Sequential(nn.Linear(16, 8),  
                      nn.Linear(8, 4),  
                      nn.Linear(4, 2))
```

```
prediction = model(sample)
```

Calculate the loss and gradients

```
criterion = CrossEntropyLoss()  
loss = criterion(prediction, target)  
loss.backward()
```

Access each layer's gradients

```
model[0].weight.grad  
model[0].bias.grad  
model[1].weight.grad  
model[1].bias.grad  
model[2].weight.grad  
model[2].bias.grad
```

Updating model parameters manually

```
# Learning rate is typically small
lr = 0.001

# Update the weights
weight = model[0].weight
weight_grad = model[0].weight.grad

weight = weight - lr * weight_grad

# Update the biases
bias = model[0].bias
bias_grad = model[0].bias.grad
bias = bias - lr * bias_grad
```

- Access each layer gradient
- Multiply by the learning rate
- Subtract this product from the weight

Gradient descent

- For non-convex functions, we will use **gradient descent**
- PyTorch simplifies this with **optimizers**
 - Stochastic gradient descent (SGD)

```
import torch.optim as optim

# Create the optimizer
optimizer = optim.SGD(model.parameters(), lr=0.001)

# Perform parameter updates
optimizer.step()
```


Let's practice!

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